The Journal of Physiology

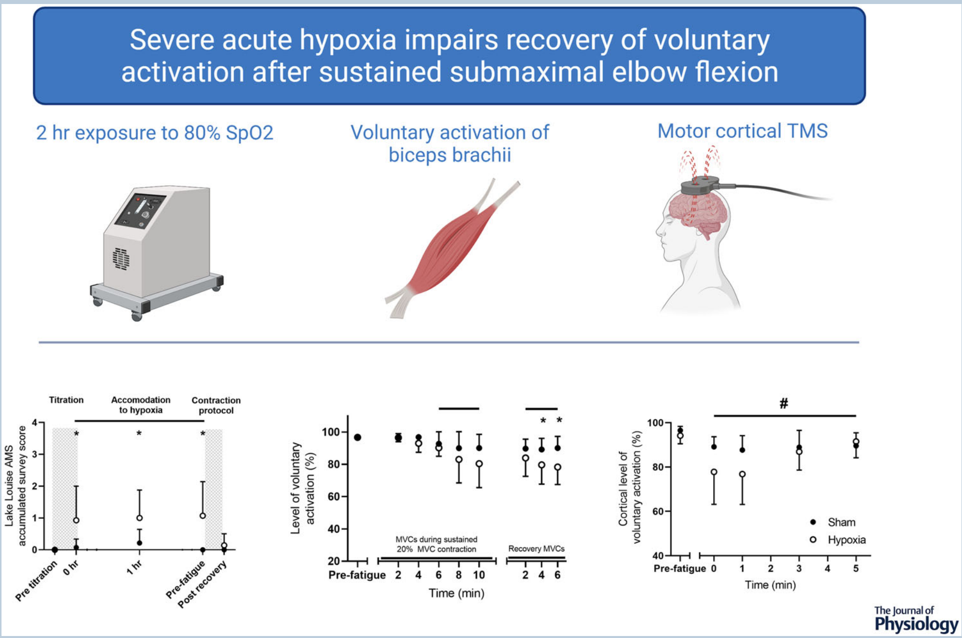
Severe acute hypoxia impairs recovery of voluntary muscle activation after sustained submaximal elbow flexion

Daniel J. McKeown[1], Chris J. McNeil[2], Emily J. Brotherton[1], Michael J. Simmonds[3] and Justin J. Kavanagh[1]

[1]*Neural Control of Movement Laboratory, Menzies Health Institute Queensland, Griffith University, Gold Coast, Queensland, Australia*
[2]*Integrated Neuromuscular Physiology Laboratory, Centre for Heart, Lung, and Vascular Health, School of Health and Exercise Sciences, University of British Columbia, Kelowna, British Columbia, Canada*
[3]*Biorheology Research Laboratory, Menzies Health Institute Queensland, Griffith University, Gold Coast, Queensland, Australia*

Edited by: Scott Powers & Richard Carson

The peer review history is available in the Supporting Information section of this article (https://doi.org/10.1113/JP281897#support-information-section).



Abstract The purpose of this study was to determine how severe acute hypoxia alters neural mechanisms during, and following, a sustained fatiguing contraction. Fifteen participants (25 ± 3.2 years, six female) were exposed to a sham condition and a hypoxia condition where they performed a 10 min elbow flexor contraction at 20% of maximal torque. For hypoxia, peripheral blood oxygen saturation (S_{pO_2}) was titrated to 80% over a 15 min period and maintained

Daniel McKeown obtained a BExSc Hons at Griffith University (Gold Coast, Australia), where he is currently completing his PhD candidature. Daniel works in the Neural Control of Movement Laboratory (Menzies Health Institute Queensland) under the supervision of Assoc. Prof. Justin Kavanagh and Assoc. Prof. Michael Simmonds. Daniel's work focuses on how exposure to a hypoxic environment alters neural mechanisms within the brain and spinal cord that control muscle activation during exercise.

 DOI: 10.1113/JP281897

for 2 h. Maximal voluntary contraction torque, EMG root mean square, voluntary activation, rating of perceived muscle fatigue, and corticospinal excitability (motor-evoked potential) and inhibition (silent period duration) were then assessed before, during and for 6 min after the fatiguing contraction. No hypoxia-related effects were identified for neuromuscular variables during the fatigue task. However, for recovery, voluntary activation assessed by motor point stimulation of biceps brachii was lower for hypoxia than sham at 4 min (sham: 89% ± 7%; hypoxia: 80% ± 12%; $P = 0.023$) and 6 min (sham: 90% ± 7%; hypoxia: 78% ± 11%; $P = 0.040$). Similarly, voluntary activation ($P = 0.01$) and motor-evoked potential area ($P = 0.002$) in response to transcranial magnetic stimulation of the motor cortex were 10% and 11% lower during recovery for hypoxia compared to sham, respectively. Although an S_{pO_2} of 80% did not affect neural activity during the fatiguing task, motor cortical output and corticospinal excitability were reduced during recovery in the hypoxic environment. This was probably due to hypoxia-related mechanisms involving supraspinal motor circuits.

(Received 9 May 2021; accepted after revision 4 November 2021; first published online 1 November 2021)
Corresponding author Daniel J. McKeown: Neural Control of Movement Laboratory, Menzies Health Institute Queensland, School of Allied Health Sciences, Griffith University, Gold Coast, Queensland 4222, Australia.
Email: daniel.mckeown@griffith.edu.au

Abstract figure legend. Participants were exposed to 2 h of severe acute hypoxia where voluntary activation of the elbow flexors was assessed during, and following, a sustained submaximal contraction for 10 min. Participants experienced greater symptoms of acute mountain sickness throughout the 2 h exposure. Voluntary activation, measured at both the motor nerve of the biceps brachii and the motor cortex, was reduced during recovery from the sustained submaximal contraction. The impairment in voluntary activation recovery during the recovery phase was probably due to hypoxia-related mechanisms involving supraspinal motor circuits.

Key points

- Acute hypoxia has been shown to impair voluntary activation of muscle and alter the excitability of the corticospinal motor pathway during exercise.
- However, little is known about how hypoxia alters the recovery of the motor system after performing fatiguing exercise.
- Here we assessed hypoxia-related responses of motor pathways both during active contractions and during recovery from active contractions, with transcranial magnetic stimulation and motor point stimulation of the biceps brachii.
- Fatiguing exercise caused reductions in voluntary activation, which was exacerbated during recovery from a 10 min sustained elbow flexion in a hypoxic environment.
- These results suggest that reductions in blood oxygen concentration impair the ability of motor pathways in the CNS to recover from fatiguing exercise, which is probably due to hypoxia-induced mechanisms that reduce output from the motor cortex.

Introduction

The contribution that central and peripheral mechanisms play in the development of exercise-induced fatigue differs with a number of factors. These include the intensity of contraction (Allen *et al*. 1998), the duration of contraction (Bigland-Ritchie *et al*. 1986; Sogaard *et al*. 2006) and oxygen availability (Amann *et al*. 2006). Numerous studies have reported that prolonged submaximal contractions induce a progressive increase in fatigue (indicated by a reduction in torque output during brief maximal voluntary and electrically evoked contractions), which aligns with a progressive increase in neural drive to the fatiguing muscle to sustain the force being generated (Bigland-Ritchie *et al*. 1986; Hunter *et al*. 2002; Adam & De Luca, 2003, 2005; Sogaard *et al*. 2006; McNeil *et al*. 2011a). While it might be expected that neural drive to the fatiguing muscle would be consistently, and predictably, impaired under conditions of reduced blood oxygen concentrations, the evidence surrounding this is remarkably diverse. Indeed, several studies that have examined submaximal contractions performed in an acute hypoxic environment report impaired motor performance and exacerbated supraspinal fatigue (Dousset *et al*. 2001; Katayama *et al*. 2007; Millet *et al*. 2012; Ruggiero *et al*. 2018), whereas other studies report few hypoxia-related effects (Rupp *et al*. 2015; Jacunski & Rafferty, 2020).

Under normoxic conditions, the decline in force that is typically observed during fatiguing contractions recovers quickly within 3–4 min and then more slowly over the next 10–20 min (Sogaard *et al*. 2006; Froyd *et al*. 2013; Heroux *et al*. 2016). It is possible that a differential recovery of central and peripheral components of fatigue contribute to the time course of this response, as the superimposed twitch and silent period duration during maximal voluntary contractions (MVCs; central) substantially recover within a few minutes of performing a sustained 15% MVC elbow flexion, but resting twitch (peripheral) exhibits minimal recovery for up to 25 min (Sogaard *et al*. 2006). Although there is considerable evidence that acute hypoxia exacerbates the impairment of voluntary activation immediately after a fatiguing task (Goodall *et al*. 2010; Rupp *et al*. 2015; Ruggiero *et al*. 2018), it surprising that effects do not seem to persist during the recovery phase (Katayama *et al*. 2007; Szubski *et al*. 2007; Rupp *et al*. 2015; Townsend *et al*. 2021). However, it should be noted that these recovery studies allowed for 6–10 min of recovery time to pass before assessing voluntary activation (Katayama *et al*. 2007; Rupp *et al*. 2015; Townsend *et al*. 2021), or returned the participant to a normoxic environment during the recovery period (Katayama *et al*. 2007; Townsend *et al*. 2021). As such, our understanding of how oxygen availability influences the recovery phase of exercise may benefit by obtaining recovery measurements closer to the cessation of exercise with a hypoxic stimulus that is consistent with the exercise task.

The purpose of the present study was to determine if exposure to severe acute hypoxia alters the development, and recovery, of neuromuscular fatigue due to a sustained submaximal contraction in humans. The peripheral blood oxygen saturation (S_{pO_2}) of participants was titrated to 80% over a period of 15 min, and remained at this level for the duration of testing. After 2 h of exposure to 80% S_{pO_2}, a 20% MVC isometric elbow flexion was performed for 10 min. Transcranial magnetic stimulation (TMS) and motor point stimulation were used to assess the central and peripheral contributions to fatigue. It was hypothesized that MVC torque and central mechanisms contributing to voluntary activation of the elbow flexors would be reduced due to the contraction protocol, and this reduction would be exacerbated under conditions of hypoxia compared to sham. It was also hypothesized that voluntary activation of the elbow flexors would begin to recover rapidly after the sustained submaximal contraction, although recovery of voluntary activation would be delayed under conditions of hypoxia compared to sham.

Materials and method

Ethical approval

Each participant provided written and witnessed informed consent prior to undertaking testing. All experimental procedures were approved by the Griffith University Human Research Ethics Committee (reference number: 2018/838). The study conformed to the standards set by the *Declaration of Helsinki*, except for registration in a database.

Participants

Fifteen heathy individuals (25 ± 3.2 years, six female) volunteered to participate in the current study. All participants completed a medical history questionnaire prior to testing, which included exclusion criteria specific to reduced blood oxygenation, magnetic stimulation and electrical stimulation. Participants were not permitted to take any form of CNS medications and were instructed to refrain from any form of stimulant or depressant such as caffeine, alcohol or moderate- to high-intensity exercise for 12 h before testing. One female participant withdrew from the study after experiencing syncope during the hypoxia titration phase. Power calculations were performed with G∗Power software (v3.1.9.7). A statistical power of 0.9 was achieved by the inclusion of 15 participants for this study at an alpha level of 0.05 and effect size of 0.25.

Experimental design

The current study was a two-way, sham-controlled, cross-over design in which participants attended two testing sessions that were separated by at least 1 week. For safety reasons this was a single-blind study, as an investigator in the room needed knowledge of the intervention and monitored S_{pO_2} throughout the testing sessions. In one session, a hypoxia intervention was employed whereas in the other session, a sham intervention was employed. The sham intervention reflected all hypoxia testing procedures except for reduction of S_{pO_2} (Fig. 1). The administration of the intervention was counterbalanced to avoid order effects, where eight of the participants underwent the hypoxia intervention in their first session, and seven of the participants underwent the sham intervention in their first session.

Instrumentation

Hypoxia intervention. An altitude simulator (ATS200HP; Altitude Training Systems, Arndell Park, NSW, Australia) and a sealed facemask was used to reduce S_{pO_2} to 80% via regulation of the fraction of inspired oxygen (F_{IO_2}). As sensitivity to altered oxygen availability is quite variable between individuals (Rojas-Camayo *et al.* 2018), F_{IO_2} was titrated over a 15 min period for each individual at a rate of 0.01 F_{IO_2} every 1.5 min and remained at the desired F_{IO_2} that consequently resulted in 80% S_{pO_2} for the 2 h experimental protocol. This hypoxic stimulus was chosen as previous research has documented hypoxia-induced changes in neuromuscular fatigue at similar S_{pO_2} levels (75–80%; Goodall *et al.* 2010, 2012; Ruggiero *et al.* 2018) and durations of exposure (2–3 h; Rupp *et al.* 2012; McKeown *et al.* 2019, 2020). S_{pO_2} and heart rate were monitored at all times using a pulse oximeter (Model 7500; Nonin Medical Inc., Plymouth, MN, USA) attached to the middle finger of the left hand. Participants were carefully monitored for symptoms of acute mountain sickness (AMS), including dizziness, headaches, sleepiness and nausea, throughout the experimental protocol. Participants scored their experience of each individual symptom as either 0 (absent), 1 (mild), 2 (moderate) or 3 (severe). Symptoms were then calculated as an accumulative score to represent the severity of AMS symptoms experienced by a participant at a time point. The Lake Louise AMS survey was employed before titration (pre), and immediately after (0 h), 1 h after and 2 h after titration before the contraction protocol (pre-fatigue), and after recovery of the contraction protocol (post-fatigue).

Electromyography and force. Participants sat in a chair with their right arm fixed in a custom-built force transducer to measure isometric elbow flexion force, which was later converted to torque offline (Fig. 1*A*). The shoulder and elbow were placed in 90° of flexion and the participant's arm was fixed to the device at the wrist (McNeil *et al.* 2011*b*; Kavanagh *et al.* 2019). A precision S-beam load cell (PT4000, PT Ltd, Auckland, New Zealand) with a 1.1 kN range and full-scale output of 3 mV/V was used to measure elbow flexion torque. Surface EMG was recorded from biceps brachii and triceps brachii by attaching circular 24 mm Ag/AgCl electrodes (Kendall Arbo; Cardinal Health Inc., Dublin, OH, USA) on each muscle in a bipolar arrangement (24 mm interelectrode distance). A ground electrode was placed on the acromion of the test limb. Torque and EMG signals were sampled at 2000 Hz using a 16-bit analog-to-digital converter (CED 1401; Cambridge Electronic Design, Cambridge, UK) and Spike2 software (version 7.02; Cambridge Electronic Design Ltd.). EMG signals were amplified (×300) and bandpass filtered (10-1000 Hz) using a CED 1902 amplifier (Cambridge Electronic Design Ltd.), whereas torque signals remained unfiltered.

Motor point stimulation. Intramuscular nerve fibres of the biceps brachii were stimulated with single supramaximal pulses of 100 μs duration via a constant current stimulator (DS7AH; Digitimer Ltd, Welwyn Garden City, Hertfordshire, UK). A surface anode was positioned over the distal biceps brachii tendon and a surface cathode was positioned over the muscle belly of the biceps brachii. Optimal positioning of the surface cathode was determined by using a motor point stimulation pen (at a stimulus intensity of 50 mA), and was based on the highest elicited elbow flexor twitch torque. Maximal resting twitch was determined by progressively increasing the stimulator current until there was no longer an increase in the elicited elbow flexor twitch torque. The stimulus intensity was

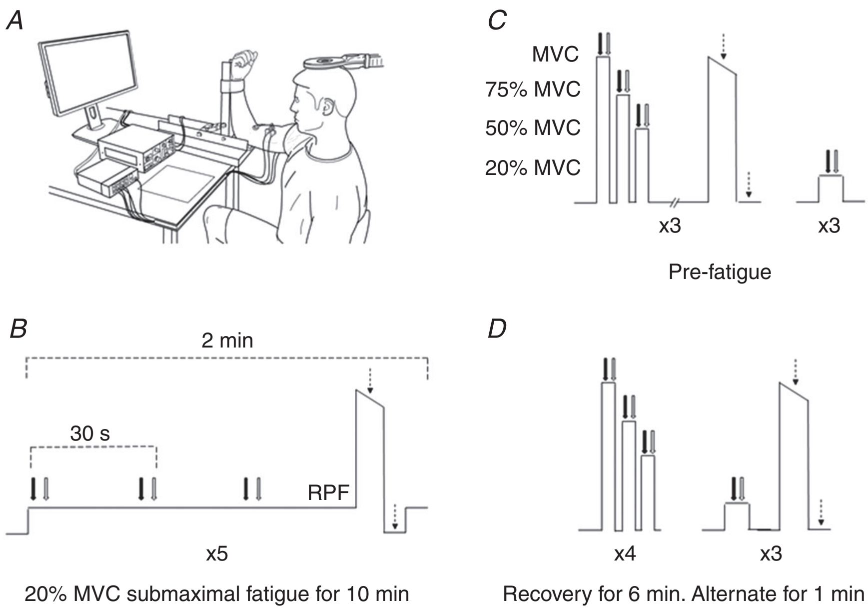


Figure 1. Contraction protocol following 2 h of sham or hypoxia
Participants attended the laboratory on two separate occasions where they underwent a 15 min titration phase. Participants sat in a chair with their right arm fixed in a custom-built force transducer to measure isometric elbow flexion torque (*A*). After 2 h of sham or hypoxia, the pre-fatigue contraction protocol (*B*) was completed, followed by the sustained submaximal contraction protocol (*C*) and recovery phase (*D*). A week later, participants completed the experimental protocol during the alternative condition. Black arrows represent transcranial magnetic stimulation, grey arrows represent brachial plexus stimulation, dashed arrows represent motor point stimulation, and rating of perceived muscle fatigue represents rate of perceived muscle fatigue assessment.

then set at 20% above the current that elicited the maximal resting twitch (sham protocol: 210 ± 83 mA, 96–360 mA; hypoxia protocol: 207 ± 82 mA, 96–360 mA). There were no significant differences in stimulus intensity between the sham and hypoxia protocols ($P = 0.640$).

Brachial plexus stimulation. A second constant current stimulator (DS7AH; Digitimer Ltd) was used to deliver single electrical pulses of 100 μs duration to the brachial plexus (Erb's point) to determine the maximal compound action potential (Mmax) of the biceps and triceps brachii. A surface anode was positioned over the ipsilateral acromion and a surface cathode was positioned over the supraclavicular fossa. Positioning of the cathode was again determined by using a motor point stimulating pen (at a stimulus intensity of 20 mA). Optimal stimulus intensity was determined by progressively increasing the stimulator current until the Mmax of both the biceps and triceps brachii was reached. The stimulus intensity was then set at 30% above that used to obtain Mmax during the sham protocol (sham protocol: 176 ± 29 mA, 130–234 mA; hypoxia protocol 164 ± 42 mA, 78–234 mA). There were no significant differences in stimulus intensity between the sham and hypoxia protocols ($P = 0.080$)

Cortical stimulation. A Magstim 200^2 TMS unit (Magstim Co Ltd., Spring Gardens, Whitland, UK) was used to elicit motor evoked potentials in the test limb. A circular coil (90 mm Remote Coil, Magstim Co Ltd.) was positioned over the vertex to elicit motor evoked potentials in the biceps and triceps brachii muscles. The coil was orientated to preferentially activate the left motor cortex projecting to the participant's right arm. The stimulator intensity which elicited the highest elbow flexor twitch torque, with a motor evoked potential that was >80% of the biceps brachii Mmax and <20% of the triceps brachii Mmax, during a brief isometric 50% MVC was chosen as the optimal stimulus intensity for each testing session (70–75% stimulus output). This criterion ensured that the TMS pulse activated a large proportion of the biceps brachii motoneuron pool whilst minimizing unintended activation of the antagonist muscle (triceps brachii; Todd *et al*. 2016).

Ratings of perceived muscle fatigue. Ratings of perceived muscle fatigue were obtained using a CR-10 Borg scale (Borg, 1982) which was administered at 2 min intervals during the 10 min sustained submaximal contraction task. Participants were informed that the CR-10 Borg scale was a measure of their subjective experience of muscle fatigue (i.e. how fatigued their biceps brachii muscle felt), regardless of objective measures of fatigue (i.e. their performance during the contraction task). Before each testing session, participants were provided a printout of the CR-10 Borg scale. Participants were instructed that the lowest value (0) on the scale represented 'no fatigue at all, where you feel that you are able to fully contract your muscle', and the highest value (10) on the scale represented 'maximal fatigue, where you feel that you are unable to contract your muscle'.

Experimental procedures

Participants visited the laboratory on two occasions, where they were exposed to a sham or hypoxia intervention that titrated S_{pO_2} to 80% over a 15 min period. Measurements were performed during a pre-fatigue contraction protocol, during the 10 min sustained submaximal contraction task, and during the 6 min recovery phase (Fig. 1). Participants were provided with visual feedback of instantaneous torque output via a computer monitor that was placed at eye-level, ~60 cm in front of them. For maximal contractions, participants were encouraged to pull on the force transducer maximally for the duration of the contraction. After each MVC, automated target lines were provided on the computer monitor at 75, 50 and 20% MVC torque. For these submaximal contractions, participants were required to match their torque output with the target to the best of their ability.

Pre-fatigue contraction protocol. Before commencing the sustained submaximal contraction task, six sets of pre-fatigue MVCs as well as three sets of submaximal pre-fatigue contractions were performed (Fig. 1*B*). Three sets of the MVCs consisted of performing a sequence of three contractions at 100, 75 and 50% MVC torque. Three seconds of rest was given between each contraction to allow for accurate targeting of the desired contraction intensity. During each of these contractions, TMS was applied to the motor cortex, and electrical stimulation was applied to the brachial plexus 2 s later. The other three sets of MVCs consisted of a brief MVC, with a motor point stimulation delivered to the biceps brachii during and ~1 s after the MVC. Participants were given 2 min of rest between each of the pre-fatigue MVCs and the two types of MVC sets were performed in alternating order. The highest peak torque of the six MVCs was used to calculate the 20% MVC target trace for the sustained submaximal contraction task. Participants were then instructed to briefly hold the 20% MVC torque target, with TMS applied to the motor cortex and electrical stimulation was applied to the brachial plexus 2 s later. Thirty seconds of rest was given between each of the three submaximal pre-fatigue contractions.

Sustained submaximal contraction task. This protocol consisted of a sustained submaximal elbow flexion at 20% of the peak pre-fatigue MVC torque, while performing a brief maximal contraction every 2 min, for a period of 10 min (Fig. 1*C*). TMS and brachial plexus stimulation were applied at 30 s intervals for the duration of the sustained submaximal contraction task. During the brief MVCs, motor point stimulation was applied at peak torque and then during a brief rest. Participants were then required to immediately return to the 20% MVC torque target. Guide trace lines were provided at 5% greater and 5% less than the target. A contraction intensity of 20% MVC was selected to mitigate the effect of local muscle perfusion disruptions that occur secondary to the large extravascular pressures generated during higher intensity contractions (McNeil *et al*. 2015).

Recovery phase. The recovery phase (Fig. 1*D*) consisted of two types of contraction sets. The first type involved a sequence of brief contractions at 100, 75 and 50% MVC torque (submaximal targets were scaled to the peak torque of the MVC), with TMS and brachial plexus stimulation delivered during each contraction. The second type began with a brief contraction at 20% of pre-fatigue MVC torque (i.e. the target for the fatiguing task) that included TMS and brachial plexus stimulation, followed by a brief MVC with motor point stimulation during and after the contraction. Sets of the first type were performed immediately as well as 1, 3 and 5 min after the fatigue protocol, whereas the second type was performed at 2, 4 and 6 min of recovery.

Data analysis

All torque and electrophysiology data were analysed offline using Spike2 (version 7.02; Cambridge Electronic Design Ltd.). Scores from the Lake Louise AMS survey are reported as an accumulative score, and ratings of perceived muscle fatigue measurements are reported as change scores from 2 min of the sustained submaximal contraction task. Torque measurements, root mean square of EMG (EMG_{RMS}), and biceps brachii silent period duration data during the fatigue protocol and recovery phase were expressed relative to pre-fatigue measurements.

MVC variables. MVC torque generation and EMG_{RMS} of the biceps brachii were calculated during a 100 ms window before the stimulus artifact of the six pre-fatigue MVCs, the five MVCs during the sustained submaximal contraction task and the seven MVCs during the recovery phase. Peak torques of motor point superimposed and resting twitches were calculated from increases in torque output following electrical stimulation. Level of voluntary activation was calculated as [1 − (superimposed twitch/resting twitch)] × 100, where the superimposed and resting twitch were associated with the same contraction. To calculate cortical level of voluntary activation, an estimated resting twitch was determined by linearly regressing TMS-elicited superimposed twitches of 100, 75 and 50% MVC with relative voluntary torque (R^2 values ranged from 0.90 to 1.0). The y-intercept of the regression was used to estimate a resting twitch. The above-mentioned calculation was then used to calculate cortical level of voluntary activation.

20% MVC variables. Torque generation and EMG_{RMS} of the biceps brachii were calculated during a 100 ms window prior to the TMS stimulus artifact during pre-fatigue contractions, during the sustained submaximal contraction task and during the 20% MVCs of the recovery phase. EMG_{RMS} of the biceps brachii was normalized to the Mmax peak-to-peak amplitude during the same contraction. Peak torques of TMS superimposed twitches were calculated from increases in torque output following magnetic stimulation. Motor-evoked potential area was measured from each TMS pulse and then normalized to the area of the Mmax elicited 2 s later, to account for activity-dependent changes in muscle fibre action potentials. Silent period duration was the time from the delivery of the TMS pulse (EMG stimulus artifact) to the continuous recommencement of voluntary EMG.

Statistical analysis

Normality of the data was assessed using Shapiro–Wilk tests. Greenhouse–Geisser corrections were applied when assumptions of sphericity were violated. Differences between pre-fatigue measurements in the sham and hypoxia conditions were assessed using Student's paired t tests. To address our hypotheses, the sustained submaximal contraction task and the recovery phase were analysed separately. Two-way repeated measures ANOVAs were used to examine a main effect of condition (hypoxia *vs*. sham) and a main effect of time (10 min of measurements during the contraction task, or 6 min of measurements in the recovery phase). Where a significant main effect of time was identified, Dunnett's multiple comparisons tests were used to identify at which timepoint each variable differed from pre-fatigue measurements. Condition × time interaction effects were also examined for each dependent variable, where Bonferroni multiple comparisons tests were used to identify the timepoint(s) where hypoxia differed from the sham condition in the event of a significant interaction. Correlations between the decline in MVC torque, motor point level of voluntary activation and ratings of perceived muscle fatigue were examined using Pearson's correlations analysis. All statistical procedures were performed using IBM SPSS Statistics (v.26) with α levels set at <0.05. Effect sizes are presented as partial eta-squared values ($\eta_p{}^2$). Data in tables and figures are presented as group means and error bars represent the standard deviation of the mean.

Table 1. Torque and EMG measures for the unfatigued muscle following 2 h of sham or hypoxic exposure

	Sham	Hypoxia
Pre-fatigue MVC		
Elbow flexion torque (N.m)	69.9 ± 31.5	70.7 ± 29.6
Motor point superimposed twitch torque (N.m)	0.66 ± 0.48	0.52 ± 0.36
Motor point resting twitch torque (N.m)	15.2 ± 7.5	15.2 ± 6.8
Motor point level of voluntary activation (%)	95.7 ± 2.1	96.6 ± 1.7
Cortical level of voluntary activation (%)	96.0 ± 1.9	93.1 ± 6.1
Pre-fatigue 20% MVC		
Elbow flexion torque (N.m)	15.2 ± 7.0	15.1 ± 6.3
Biceps brachii EMG_{RMS} (% Mmax)	1.1 ± 0.6	1.3 ± 1.0
TMS-elicited superimposed twitch torque (N.m)	6.6 ± 4.0	7.7 ± 4.1
Motor-evoked potential area (% Mmax)	59.0 ± 23.2	49.3 ± 23.1
Biceps brachii Mmax area (mV/s)	0.07 ± 0.04	0.09 ± 0.05
Biceps brachii silent period (ms)	181.1 ± 50.2	145.8 ± 74.3

Data are presented as mean ± SD (n = 14).
MVC, maximal voluntary contraction; Mmax, maximal compound action potential; TMS, transcranial magnetic stimulation.

Results

Pre-fatigue neuromuscular function

All participants desaturated to 80% S_{pO_2} by the end of the titration phase. Specifically, S_{pO_2} for hypoxia was 80.8 ± 5.6% after 15 min of titration, whereas S_{pO_2} for sham was 96.4 ± 1.6% after the 15 min period. Neuromuscular function was then assessed in the unfatigued muscle following 2 h of exposure to the sham or hypoxic intervention (Table 1). No significant hypoxia-related differences were observed for any variable ($P > 0.05$), which indicates that 2 h of hypoxic exposure did not affect neuromuscular function of the unfatigued muscle.

MVC torque and EMG_{RMS} during the sustained submaximal elbow flexion

After 2 h of exposure to the sham or hypoxia intervention, MVCs were performed at 2 min intervals during the sustained submaximal contraction. No main effect of condition ($P = 0.738$, $\eta_p^2 = 0.012$) or condition × time interaction ($P = 0.981$, $\eta_p^2 = 0.089$) was detected for MVC torque (Fig. 2*A*). However, a main effect of time was detected (F (1.77, 17.69) = 28.115, $P < 0.0001$, $\eta_p^2 = 0.738$) where Dunnett's *post hoc* testing revealed MVC torque was significantly lower than pre-fatigue measurements for all MVCs during the contraction phase (all $P < 0.05$). Similarly, for biceps brachii EMG, no main effect of condition ($P = 0.553$, $\eta_p^2 = 0.028$) or condition × time interaction ($P = 0.613$, $\eta_p^2 = 0.042$) was detected (Fig. 2*C*). A main effect of time was detected (F (2.39, 31.12) = 3.791, $P = 0.027$, $\eta_p^2 = 0.226$), where Dunnett's *post hoc* testing revealed biceps brachii EMG_{RMS} was significantly greater than pre-fatigue measurements only at 2 min ($P = 0.025$).

MVC torque and EMG_{RMS} during recovery from the sustained submaximal elbow flexion

To characterize the recovery phase, MVCs were assessed immediately after the submaximal contraction and at 1 min intervals for 6 min. Similar to the contraction phase, no main effect of condition ($P = 0.175$, $\eta_p^2 = 0.245$) or condition × time interaction ($P = 0.824$, $\eta_p^2 = 0.068$) was detected for MVC torque during the recovery phase (Fig. 2*A*). However, a main effect of time was detected (F (2.08, 14.57) = 21.324, $P < 0.0001$, $\eta_p^2 = 0.753$) where Dunnett's *post hoc* testing revealed MVC torque was significantly lower than pre-fatigue measurements for all MVCs during the recovery phase (all $P < 0.05$). No main effect of condition ($P = 0.670$, $\eta_p^2 = 0.039$), time ($P = 0.318$, $\eta_p^2 = 0.196$) or condition × time interaction ($P = 0.448$, $\eta_p^2 = 0.167$) was detected for biceps brachii EMG_{RMS} (Fig. 2*C*).

Motor point voluntary activation during the sustained submaximal elbow flexion

After 2 h of exposure to the sham or hypoxia intervention, motor point stimulation was delivered during and after the brief MVCs to establish voluntary activation. No main effect of condition or condition × time interaction was detected for motor point superimposed twitch (all $P > 0.195$, $\eta_p^2 > 0.118$; Fig. 3*A*), resting twitch (all $P > 0.257$, $\eta_p^2 > 0.132$; Fig. 3*C*) or the subsequent voluntary activation calculation (all $P > 0.147$, $\eta_p^2 > 0.143$; Fig. 3*E*). A main effect of time was detected for resting twitch (F (1.20, 8.43) = 7.843, $P = 0.019$, $\eta_p^2 = 0.528$), superimposed twitch (F (1.366, 9.56) = 4.711, $P = 0.048$, $\eta_p^2 = 0.402$) and voluntary activation (F (2.91, 70.86) = 21.14, $P < 0.0001$, $\eta_p^2 = 0.490$). Dunnett's *post hoc* testing revealed that resting twitch was significantly lower than the pre-fatigue measurement throughout the sustained contraction (all $P < 0.05$), superimposed twitch was significantly greater than pre-fatigue only at 10 min ($P = 0.036$) and voluntary activation was significantly lower than pre-fatigue from 6 min onwards (all $P < 0.01$).

Motor point voluntary activation during recovery from the sustained submaximal elbow flexion

A main effect of condition was detected for motor point superimposed twitch (F (1, 9) = 7.309, P = 0.024, η_p^2 = 0.448, Fig. 3*A*), where twitch amplitude was greater for the hypoxia condition compared to the sham condition. This aligned with a significant hypoxia-related effect in voluntary activation (F (1, 9) = 5.22, P = 0.032, η_p^2 = 0.415, Fig. 3*E*), where voluntary activation was lower for the hypoxia condition compared to the sham condition. A condition × time interaction effect was also identified for superimposed twitch (F (3, 27) = 3.289, P = 0.036, η_p^2 = 0.268) and voluntary activation (F (3, 27) = 3.676, P = 0.024, η_p^2 = 0.288). Bonferroni *post hoc* tests indicated that superimposed twitch during hypoxia was larger than sham at 4 min (P = 0.006) and 6 min (P = 0.024), which aligned with voluntary activation being significantly lower for the hypoxia condition than the sham condition at 4 min (P = 0.023) and 6 min (P = 0.040). A main effect of time was detected for resting twitch (F (3, 39) = 9.979, P = 0.022, η_p^2 = 0.449, Fig. 3*C*), superimposed twitch (F (3, 27) = 7.750, P = 0.001, η_p^2 = 0.463) and voluntary activation (F(3.433, 77.68) = 15.78, $P < 0.0001$, η_p^2 = 0.650) during the recovery phase. Dunnett's *post hoc* testing revealed that the superimposed twitch was significantly greater than pre-fatigue (all $P < 0.05$), whereas the resting twitch (all $P < 0.01$) and voluntary activation (all $P < 0.05$) were significantly lower than pre-fatigue throughout the recovery phase.

Correlation between decrease in MVC torque, motor point voluntary activation and rating of perceived muscle fatigue during the contraction protocol

Pearson's correlation analysis was used to assess the relationship between the decline in MVC torque, motor point level of voluntary activation score, and rating of perceived muscle fatigue during the final minute of the sustained contraction (10 min), and 4 and 6 min during the recovery phase (Table 2). A positive and significant correlation was seen between the decrease in MVC torque and voluntary activation during both the sham and hypoxia conditions at 10 min of the sustained contraction (sham: r^2 (12) = 0.438, P = 0.019; hypoxia: r^2(12) = 0.542, P = 0.003) and 4 min (sham: r^2 (12) = 0.309, P = 0.039; hypoxia: r^2 (12) = 0.456, P = 0.008) and 6 min (sham: r^2 (12) = 0.565, P = 0.002; hypoxia: r^2(12) = 0.385, P = 0.018) of the recovery phase. However, no correlation for rating of perceived muscle fatigue was identified for either MVC torque (sham: r^2 (12) = 0.078, P = 0.334; hypoxia: r^2 (12) < 0.0001; P = 0.984) or voluntary activation (sham: r^2 (12) = 0.0001, P = 0.975; hypoxia: r^2 (12) = 0.021, P = 0.623) during the sham and hypoxia condition at the final minute of the sustained contraction.

Cortical voluntary activation during recovery from the sustained submaximal elbow flexion

Single-pulse TMS of the motor cortex was used to obtain superimposed twitches from the biceps brachii during the recovery phase, which was subsequently used to calculate cortical voluntary activation. A main effect of condition was detected for cortical

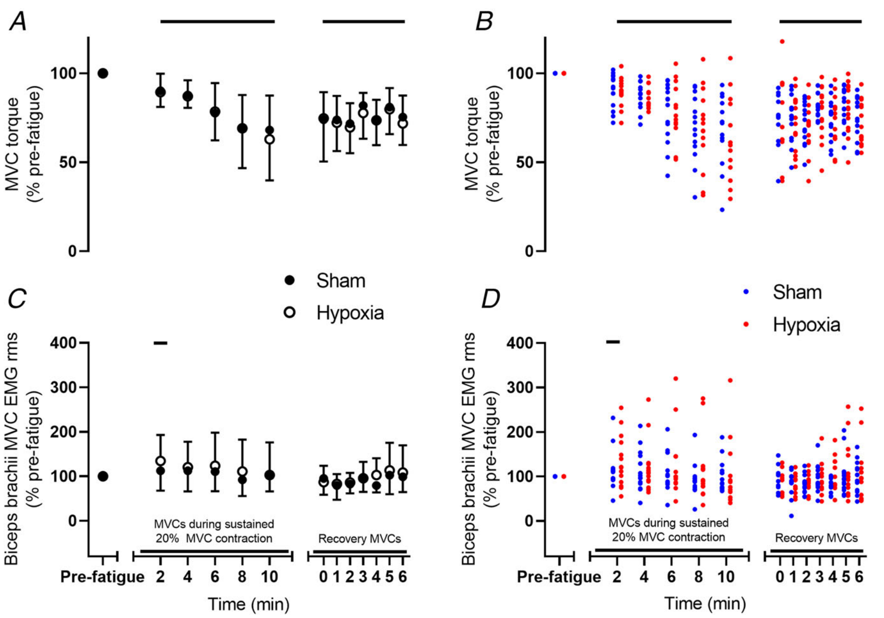


Figure 2. Elbow flexion maximal voluntary contraction (MVC) torque for the group (*A*) and individuals (*B*), and biceps brachii EMG root mean square (EMG_{RMS}) at MVC for the group (*C*) and individuals (*D*) during the sustained submaximal contraction and subsequent recovery phase
Torque and EMG data are expressed as a percentage of pre-fatigue measurements that were obtained after 2 h of hypoxic exposure or sham. A main effect of time was detected for MVC torque in both the sustained submaximal contraction and recovery, and a main effect of time was detected for biceps brachii EMG_{RMS} during the sustained submaximal contraction (solid horizontal lines represent Dunnett's tests; $P < 0.05$). Data are presented as the group mean and error bars represent the standard deviation of the mean (n = 14). [Colour figure can be viewed at wileyonlinelibrary.com]

Table 2. Correlation between reduction iN.mVC torque, voluntary activation, and rating of perceived muscle fatigue

	10 min of fatigue protocol		4 min of recovery protocol		6 min of recovery protocol	
	Pearson's r^2	*P* value	Pearson's r^2	*P* value	Pearson's r^2	*P* value
Sham						
MVC *vs*. VA	0.438	0.019*	0.309	0.039*	0.565	0.002*
MVC *vs*. RPF	0.078	0.334				
VA *vs*. RPF	0.0001	0.975				
Hypoxia						
MVC *vs*. VA	0.542	0.003*	0.456	0.008*	0.385	0.018*
MVC *vs*. RPF	<0.0001	0.984				
VA *vs*. RPF	0.021	0.623				

An asterisk denotes a significant correlation (n = 14).
MVC, reduction in maximal voluntary contraction from pre-fatigue measures; VA, motor point level of voluntary activation; RPF, rating of perceived muscle fatigue.

voluntary activation (F (1, 5) = 13.977, P = 0.013, η_p^2 = 0.737), where cortical voluntary activation was lower during the hypoxia condition compared to the sham condition (Fig. 4*A*). No condition × time interaction effect was detected for cortical voluntary activation (P = 0.791, η_p^2 = 0.022). However, a main effect of time was detected for cortical voluntary activation (F (1.51, 7.55) = 6.847, P = 0.025, η_p^2 = 0.578), where Dunnett's *post hoc* testing revealed that cortical voluntary activation was significantly lower than pre-fatigue measurements throughout recovery for sham and hypoxia (all $P < 0.05$).

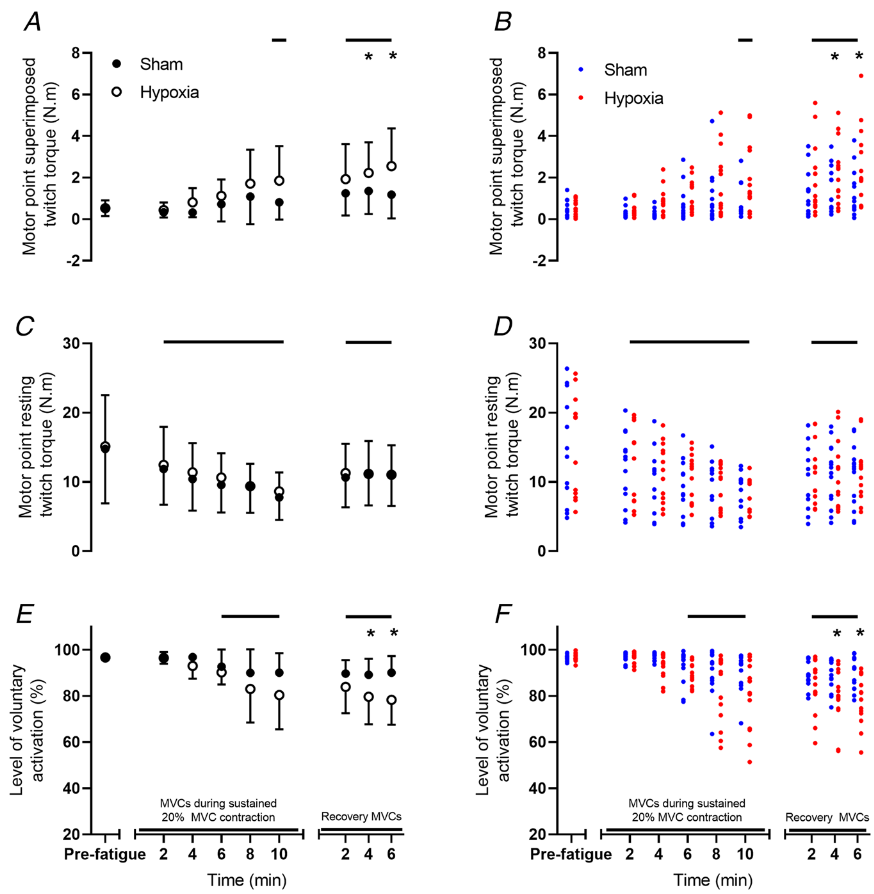


Figure 3. Motor point superimposed twitch torque for the group (*A*) and individuals (*B*), resting twitch torque for the group (*C*) and individuals (*D*), and voluntary activation for the group (*E*) and individuals (*F*)
Main effects of time were detected for resting twitch, superimposed twitch and voluntary activation during the sustained submaximal contraction task (solid horizontal lines represent Dunnett's tests; $P < 0.05$). During the recovery phase, main effects of time, condition and condition × time interaction effects were detected for superimposed twitch torque and voluntary activation (solid horizontal lines represent Dunnett's tests, and asterisks represent Bonferroni's tests for the interaction; $P < 0.05$). Data represent the group mean and error bars represent the standard deviation of the mean (n = 14). [Colour figure can be viewed at wileyonlinelibrary.com]

EMG_{RMS}, corticospinal excitability and corticospinal inhibition assessed during the sustained submaximal elbow flexion

EMG_{RMS}, motor-evoked potential area and silent period duration were assessed at 30 s intervals during the contraction task, when participants were generating 20% MVC torque. Although no main effect of condition (P = 0.252, $\eta_p{}^2$ = 0.152) or condition × time interaction (P = 0.543, $\eta_p{}^2$ = 0.028) was detected for biceps brachii EMG_{RMS}, a main effect of time was detected for biceps brachii EMG_{RMS} (F (4.840, 62.92) = 5.257, P = 0.0005, $\eta_p{}^2$ = 0.372, Fig. 5*A*). Dunnett's *post hoc* testing revealed that biceps brachii EMG_{RMS} was significantly greater than pre-fatigue from 2.5 min onwards (all $P < 0.05$). Corticospinal excitability followed a similar pattern to EMG_{RMS}. Although no main effect of condition ($P = 0.283$, $\eta_p{}^2 = 0.162$) or condition × time interaction (P = 0.375, $\eta_p{}^2$ = 0.135) was detected for motor-evoked potential area, a main effect of time was detected for motor-evoked potential area (F (15, 105) = 7.778, $P = 0.001$, $\eta_p{}^2 = 0.526$, Fig. 5*C*). Dunnett's *post hoc* testing revealed that motor-evoked potential area was significantly greater than pre-fatigue throughout the contraction (all P < 0.05). No main effects of condition ($P = 0.433$, $\eta_p{}^2 = 0.159$) and time ($P = 0.073$, $\eta_p{}^2 = 0.300$), or a condition × time interaction effect (P = 0.783, $\eta_p{}^2$ = 0.147) were identified for silent period duration.

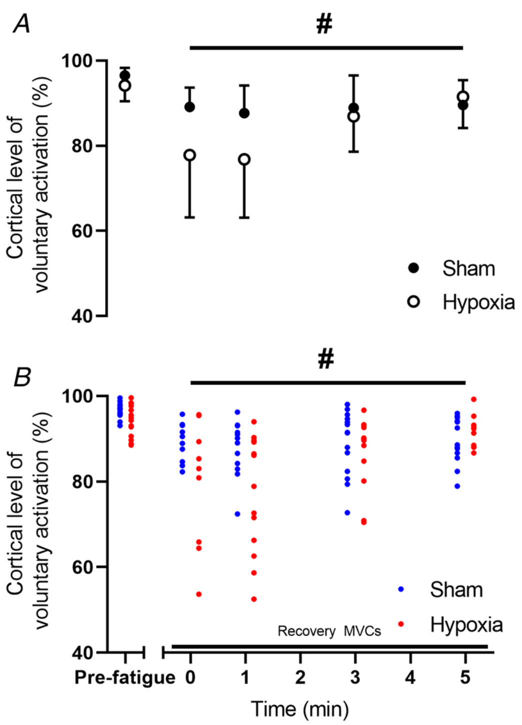


Figure 4. Cortical voluntary activation for the group (*A*) and individuals (*B*) measured pre-fatigue and throughout the recovery phase
Main effects of condition (represented by the hash symbol; P < 0.05) and time (solid horizontal lines represent Dunnett's tests; P < 0.05) were detected for cortical voluntary activation during the recovery phase, where cortical voluntary activation was lower for hypoxia than sham and lower than pre-fatigue measurements throughout recovery for both sham and hypoxia. Data represent the group mean and error bars represent the standard deviation of the mean (n = 14). [Colour figure can be viewed at wileyonlinelibrary.com]

EMG_{RMS}, corticospinal excitability and corticospinal inhibition assessed during recovery from the sustained submaximal elbow flexion

EMG_{RMS}, motor-evoked potential area and silent period duration were also assessed at 2 min intervals during the recovery phase, when participants were generating 20% MVC torque. Although no main effect of condition (P = 0.187, $\eta_p{}^2$ = 0.140) or condition × time interaction (P = 0.499, $\eta_p{}^2$ = 0.063) was detected for biceps brachii EMG_{RMS}, a main effect of time was detected for biceps brachii EMG_{RMS} (F (3, 39) = 7.556, P = 0.0004, $\eta_p{}^2$ = 0.345, Fig. 5*A*). Dunnett's *post hoc* testing revealed that EMG_{RMS} was significantly greater than pre-fatigue measurements throughout the recovery phase (P < 0.05). A main effect of condition was detected for motor-evoked potential area (F (1, 7) = 21.642, P = 0.002, $\eta_p{}^2$ = 0.756), where motor-evoked potential area was smaller for the hypoxia condition compared to the sham condition throughout the recovery (Fig. 5*C*; all P < 0.05). No main effect of time (P = 0.065, $\eta_p{}^2$ = 0.346) or a condition × time interaction effect ($P = 0.307$, $\eta_p{}^2 = 0.155$) was detected for motor-evoked potential area. No main effects of condition ($P = 0.368$, $\eta_p{}^2$ = 0.102), time (P = 0.723, $\eta_p{}^2$ = 0.053) or a condition × time interaction effect (P = 0.693, $\eta_p{}^2 = 0.058$) was identified for silent period duration.

Perception of fatigue and symptoms of hypoxia

Very mild symptoms of AMS were experienced by participants during the exposure to hypoxia (Fig. 6*A*). A main effect of time (F (2.298, 29.877) = 6.011, $P = 0.005$, $\eta_p{}^2 = 0.316$), a main effect of condition (F (1, 13) = 29.885, P = 0.001, $\eta_p{}^2$ = 0.697) and a condition × time interaction effect (F (2.295, 29.834) = 5.616, P = 0.006, $\eta_p{}^2$ = 0.302) were detected for AMS. Bonferroni *post hoc* tests identified that AMS survey scores were higher during hypoxia compared to sham at 0 h ($P = 0.005$), 1 h (P = 0.006) and 2 h (pre-fatigue; P = 0.002). Dunnett's *post hoc* testing revealed that scores were significantly higher than pretitration during hypoxia at 0 h ($P = 0.006$), 1 h (P = 0.001) and 2 h (pre-fatigue; P = 0.002). Rating of perceived muscle fatigue was measured at 2 min intervals during the sustained submaximal contraction task (Fig. 6*C*). A main effect of time was observed

(F (1.43, 18.66) = 121.192, P = 0.001, η_p^2 = 0.903). Dunnett's *post hoc* testing revealed that rating of perceived muscle fatigue was significantly greater than the initial rating from 4 min onwards (all $P < 0.001$). No main effect of condition ($P = 0.688$, $\eta_p^2 = 0.013$) or condition × time interaction (P = 0.698, η_p^2 = 0.041) was detected for rating of perceived muscle fatigue.

Discussion

The purpose of this study was to determine how severe acute hypoxia alters mechanisms that underlie the performance of a fatiguing submaximal contraction and the subsequent recovery phase. S_{pO_2} of participants was titrated to 80% and remained at this level for the duration of testing. After 2 h of exposure to 80% S_{pO_2}, a 10 min 20% MVC isometric elbow flexion was performed, where TMS and motor point stimulation were used to assess the central and peripheral contributions to fatigue. Our main findings were (1) severe acute hypoxia did not adversely affect maximal torque capacity during brief MVCs performed during, or following, the 10 min submaximal contraction, (2) voluntary activation of the elbow flexors was unaffected by hypoxia during the submaximal contraction but was significantly lower for hypoxia compared to sham following the contraction, and (3) corticospinal excitability was unaffected by hypoxia during the submaximal contraction but was lower for hypoxia than sham during the recovery phase. These findings align with other studies assessing the hypoxia-related effects on performance and supra-spinal fatigue during sustained submaximal contractions (Dousset *et al.* 2001; Katayama *et al.* 2007; Millet *et al.* 2012; Ruggiero *et al.* 2018).

Maximal elbow flexion torque and voluntary activation during a sustained submaximal contraction was unaffected by severe acute hypoxia

Fatigue is classically defined by an exercise-induced reduction in the ability to maximally produce force and is often used as an indicator of general motor performance

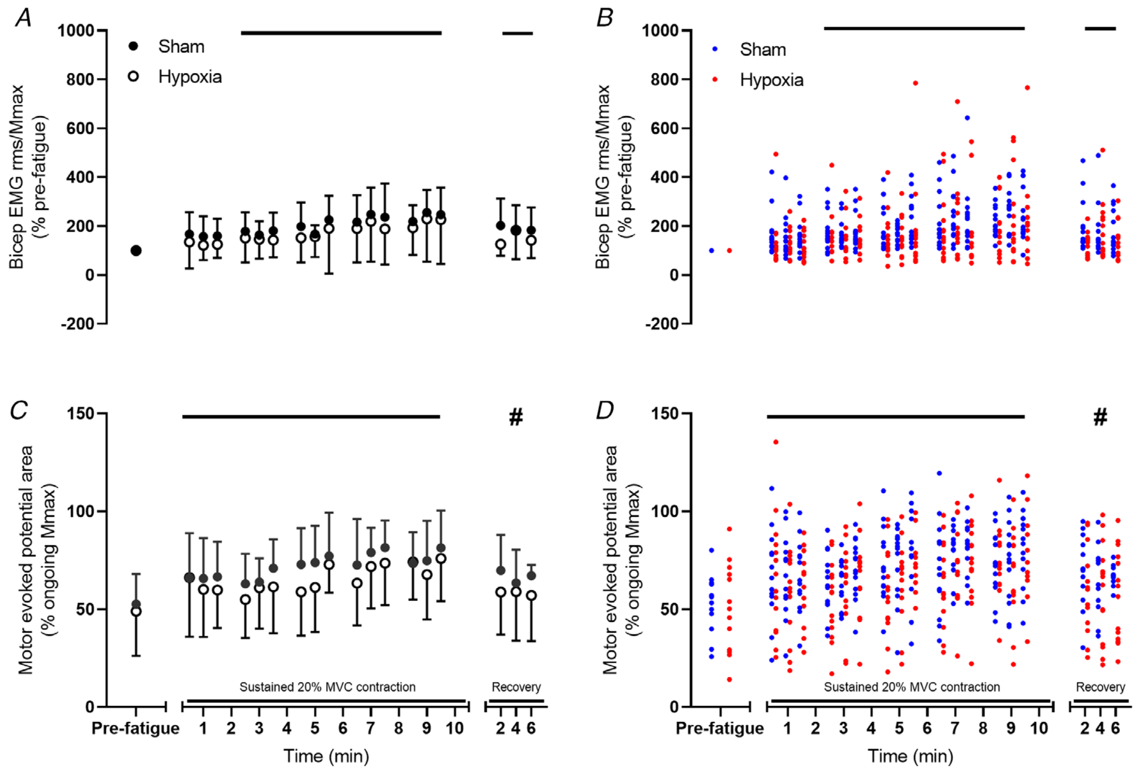


Figure 5. Biceps brachii EMG root mean square (EMG_{RMS}) for the group (*A*) and individuals (*B*), and corticospinal excitability (motor-evoked potential area) for the group (*C*) and individuals during the 20% maximal voluntary contraction (MVC) sustained submaximal contraction task and recovery phase
Main effects of time were detected for biceps brachii EMG_{RMS} and motor-evoked potential area normalized to the ongoing maximal compound action potential (Mmax; solid horizontal lines represent Dunnett's tests; $P < 0.05$) during the sustained submaximal contraction task and recovery phase. A main effect of condition was detected for motor-evoked potential area during recovery, where motor-evoked potential area was smaller during hypoxia compared to sham (represented by the hash symbol; $P < 0.05$). Data represent the group mean and error bars represent the standard deviation of the mean (n = 14). [Colour figure can be viewed at wileyonlinelibrary.com]

(Merton, 1954). The decline in MVC torque is not only due to an inability of the contractile apparatus to produce force, but also due to inadequate neural drive to the exercising muscle (Allen *et al*. 2002; Sogaard *et al*. 2006). In the current study, there was clear evidence of progressive reductions in MVC torque and central mechanisms contributing to voluntary activation of the elbow flexors, as well as a progressive reduction in resting twitch throughout the elbow flexion task. Furthermore, a moderately positive correlation was found between the reduction in MVC torque and motor point voluntary activation independent of oxygen availability. Therefore, our submaximal contraction was able to induce central fatigue, and consequently we are able to comment on the effect that severe acute hypoxia has on neural mechanisms of fatigue.

In contrast to our hypothesis, fatigue-related reductions in MVC torque and voluntary activation of the elbow flexors were not exacerbated with reduced blood oxygen concentration. However, it appears that the magnitude of reduction in MVC torque is a good predictor of motor point voluntary activation. Given that pre-fatigue measurements of MVC torque, voluntary activation and resting twitch were also unaffected by the hypoxia intervention, it appears that 2 h of exposure to 80% $S_{\mathrm{pO_2}}$ does not significantly influence central or peripheral mechanisms that contribute to activation of the unfatigued or fatigued elbow flexors. There is evidence that acute hypoxia exacerbates fatigue during dynamic knee extensions (Fulco *et al*. 1996), sustained handgrip tasks (Dousset *et al*. 2001) and intermittent, submaximal isometric contractions of the elbow flexors

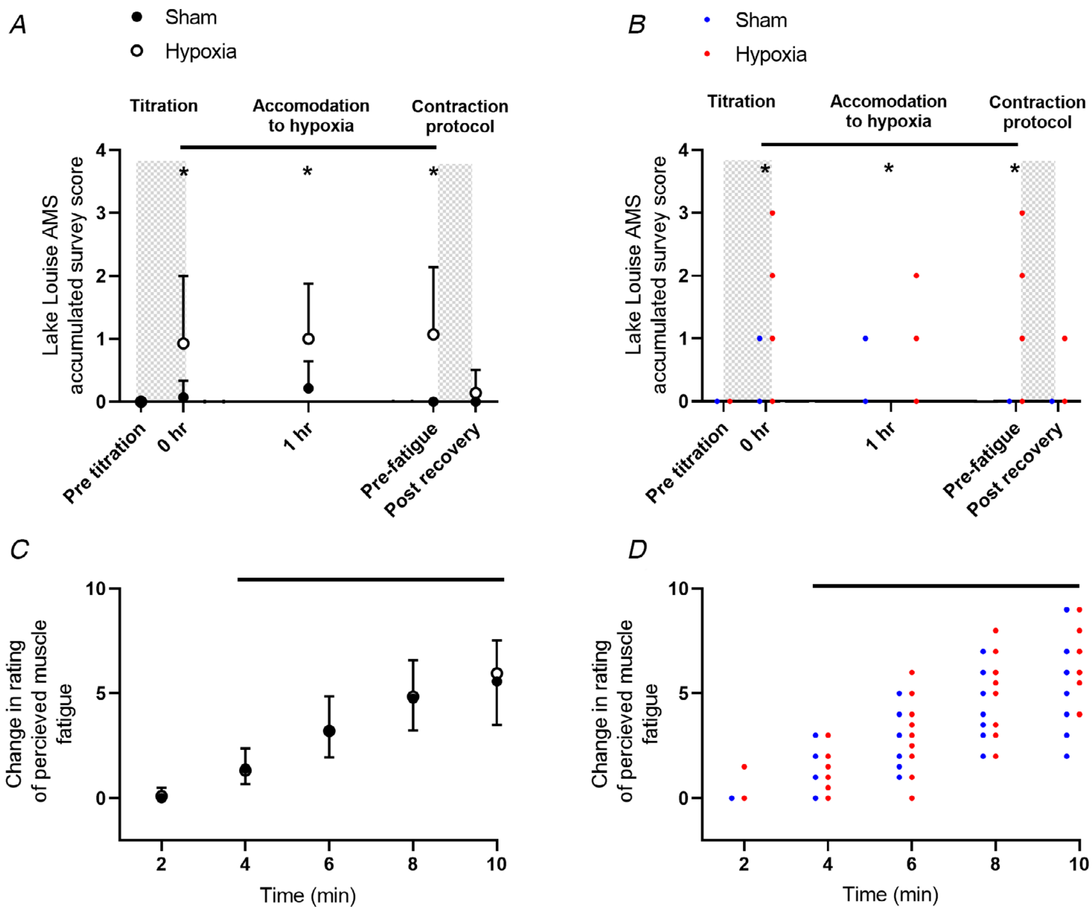


Figure 6. The Lake Louise acute mountain sickness (AMS) survey scores for the group (*A*) and individuals (*B*), and ratings of perceived muscle fatigue for the group (*C*) and individuals (*D*) measured during the submaximal contraction task

Main effects of time and condition were found for AMS. Likewise, a condition × time interaction was detected where AMS survey scores were greater during hypoxia at 0, 1 and 2 h (pre-fatigue; solid horizontal lines represent Dunnett's tests and asterisks represent Bonferroni's tests; $P < 0.05$). Grey columns denote the 15 min titration phase, and the duration of the submaximal contraction task and recovery phase. A main effect of time was observed for rating of perceived muscle fatigue, where, from 4 min onwards, ratings were greater than the rating at 2 min (solid horizontal lines represent Dunnett's tests; $P < 0.05$). Data represent the group mean and error bars represent the standard deviation of the mean ($n = 14$). [Colour figure can be viewed at wileyonlinelibrary.com]

(Millet *et al.* 2012; Ruggiero *et al.* 2018). However, the current study is consistent with experiments involving sustained submaximal isometric contractions of the soleus (Rupp *et al.* 2015), sustained maximal contractions of the first dorsal interosseous (Szubski *et al.* 2007) and sustained maximal contractions of the elbow flexors (McKeown *et al.* 2020), where acute hypoxia did not impair task performance or exacerbate fatigue during the sustained contraction task. As central fatigue takes time to develop during submaximal contractions (Sogaard *et al.* 2006), it is likely that the accumulation of central fatigue during the 10 min sustained submaximal contraction was not great enough to see hypoxia-induced impairment in voluntary activation measurements.

Hypoxia-induced impairment of voluntary activation during the recovery phase

Exercise-induced declines in voluntary activation often recover rapidly after cessation of sustained contractions under normoxic conditions. To accurately characterize this recovery, assessment of muscle function must be performed as soon as possible after termination of exercise (Gandevia *et al.* 1996; Taylor *et al.* 1996; Sogaard *et al.* 2006; Froyd *et al.* 2013; Heroux *et al.* 2016). According to the limited research assessing the recovery of voluntary activation during hypoxic exposure, impairment is not different to normoxia (Katayama *et al.* 2007; Szubski *et al.* 2007; Rupp *et al.* 2015). However, as these studies allowed substantial time to pass before assessment of voluntary activation recovery, or a return to normoxia before assessment, the hypoxia-induced effects on voluntary activation may have recovered before assessment of muscle function during these studies and may explain the conflict with our findings. In both conditions of the current study, motor point superimposed twitch, resting twitch and voluntary activation failed to return to levels that were consistent with an unfatigued muscle after 6 min of recovery. During severe acute hypoxia, motor point stimulation of the biceps brachii evoked larger superimposed twitches and lower voluntary activation compared to the sham condition. This indicates that hypoxia exacerbates the inability to recruit all motor units or discharge them at the rates needed for maximal torque. Voluntary activation scores during recovery were also lower for hypoxia than sham when superimposed twitches were evoked via TMS, which indicates that a lower S_{pO_2} reduced output from the motor cortex (Todd *et al.* 2003, 2004; Sogaard *et al.* 2006) following the sustained submaximal contraction.

Given that a hypoxia-induced impairment of voluntary activation was seen during the recovery phase, it is of interest that MVC torque did not exhibit an effect of hypoxia throughout recovery. The role of synergistic muscle activity may have contributed to preserve torque generation during the MVC efforts, which may also explain why MVC torque was also unaffected during the contraction protocol. As isometric contractions >20% MVC have been shown to blunt the delivery of oxygen-rich blood to active muscle due to increased intramuscular pressure (Barcroft & Millen, 1939; Katayama *et al.* 2010), and as the potentiated resting twitch (peripheral mechanisms) did not differ between conditions, the metabolic environment of the muscle between both the normoxic and the hypoxic conditions may have been similar.

Neural drive to biceps brachii and corticospinal excitability during sustained submaximal task

Measurements of central neural drive were made at 30 s intervals to assess how hypoxia modulates the CNS when performing a sustained submaximal contraction. As expected, EMG_{RMS} increased, suggesting an increase of motor unit recruitment and/or discharge rate to perform the task (Bigland-Ritchie *et al.* 1983; Sacco *et al.* 1997; Adam & De Luca, 2005; De Luca & Contessa, 2012). Given that this progressive increase in EMG activity did not differ for the sham and hypoxia conditions, this provides indirect evidence that hypoxia did not affect the recruitment and firing of low-threshold motor units during the 20% MVC sustained contraction.

EMG responses to TMS of the cortex were also measured at 30 s intervals during the sustained submaximal contraction task to assess the responsiveness of the corticospinal pathway during acute hypoxic exposure. Motor-evoked potential area increased progressively throughout the 10 min contraction, without any hypoxia-induced affects. Silent period duration was unchanged in either condition across the 10 min contraction. As the silent period duration represents inhibition of both cortical and spinal output (Fuhr *et al.* 1991; Di Lazzaro *et al.* 1998) and the motor-evoked potential represents motoneuronal and cortical excitability, when normalized to Mmax (Todd *et al.* 2003; Sogaard *et al.* 2006), it appears that the hypoxic exposure did not impair excitatory or inhibitory circuits during the ongoing submaximal contraction. As an effect of acute hypoxia on evoked-EMG responses has been observed during MVCs (McKeown *et al.* 2020) but not submaximal contractions (Rupp *et al.* 2015; Ruggiero *et al.* 2018; McKeown *et al.* 2020), the CNS may be required to maximally drive muscle activation for hypoxia-induced changes in corticospinal excitability to occur.

To our knowledge, this is the first study assessing corticospinal excitability during a recovery phase following fatiguing submaximal contractions in acute hypoxia. Although not in the context of fatigue, most previous research suggests that motor-evoked potential responses do not change with hypoxia (Szubski *et al.* 2006, 2007; Miscio *et al*. 2009; Goodall *et al*. 2010), whereas Rupp *et al*. (2012) showed that motor-evoked potential responses were unchanged after 1 h but increased after 3 h. During the current study, motor-evoked potential area was not affected by hypoxia during the fatiguing contraction, although the responsiveness of the corticospinal pathway was impaired during the recovery phase when exposed to severe acute hypoxia. In normoxic environments, alterations in motor-evoked potential responses have been attributed to changes in ion-channel properties, and monoaminergic and GABAergic mechanisms (Boroojerdi *et al*. 2001). As previous research has found no hypoxia-related change in motor-evoked potential responses during submaximal contraction (McKeown *et al*. 2020), which was also seen during the current study, it appears that acute severe hypoxia does not affect neuronal excitability alone. However, the present data suggest that, after repetitive activation of upper and lower motoneurons during the sustained fatiguing task, an interaction between hypoxic stimulus and prolonged motoneuron activation may exist. Although not a direct measure of motoneuron excitability, the recovery profile of H-reflex and V-wave responses did not differ after performing a fatiguing submaximal contraction task under a similar severity of acute hypoxia ($F_{IO_2} = 0.11$; Rupp *et al*. 2015). This suggests that the reduction in motor-evoked potential area may be due to hypoxia-induced impairment at a cortical level and not a spinal level. Indeed, when directly assessing motoneuron excitability in acute hypoxia, the fatigue-related reduction of cervicomedullary motor-evoked potentials was not different to that seen in normoxia, although responses were not assessed in recovery (Ruggiero *et al*. 2018). Nonetheless, as voluntary activation remained impaired during the recovery phase in hypoxia and the silent period duration was similar between both conditions, the reduction in the responsiveness of the corticospinal pathway may be due to dysfacilitation of excitatory cortical circuits responsible for muscle activation.

Perception of muscle fatigue is unimpaired during severe acute hypoxia

It is widely understood that as central motor command and descending drive to muscle increase so too does an individual's perception of effort (Enoka & Stuart, 1992; de Morree *et al*. 2012). This is due to the corollary discharge of the central motor command being processed in sensory areas of the cortex responsible for the perception of exertion (Enoka & Stuart, 1992). The CR-10 Borg scale is frequently used to assess such subjective measures of voluntary exertion during exercise (Borg, 1982). However, we used the scale to measure the participants subjective experience of muscle fatigue while performing the sustained submaximal elbow flexion task. A rating of perceived biceps brachii muscle fatigue was measured every 2 min of the sustained submaximal elbow flexion task during the sham and hypoxic conditions. Acute exposures to hypoxia have been demonstrated to exacerbate perceptions of muscle fatigue and exertion during maximal elbow flexions and exhaustive exercise (Romer *et al*. 2006; Goodall *et al*. 2014; McKeown *et al*. 2020) but failed to do so in the current study. Previous research has revealed a discrepancy in the perception of task effort and pain during a submaximal contraction task, where task effort was perceived to be greater when performed in hypoxia while elbow flexor pain remained unaffected (Ruggiero *et al*. 2018). In the current study, it is possible that the rating of perceived muscle fatigue scores reflected pain within biceps brachii more so than the effort required to maintain the contraction.

Considerations

It is well established that elbow flexion force produced by the biceps brachii muscle has a non-linear relationship with EMG (De Luca *et al*. 1982). This is often attributed to the synergistic activity of two additional muscles that contribute to elbow flexion: brachialis and brachioradialis (Murray *et al*. 2002; Bouillard *et al*. 2012). Previous research has recorded EMG in both biceps brachii and brachioradialis during fatiguing elbow flexions and have reported similar changes with fatigue (Staudenmann *et al*. 2009). However, it is possible that as the biceps brachii was accumulating fatigue during the fatigue task, elbow flexion load was distributed across additional elbow flexor muscle, which may explain the lack of difference in decline of MVC torque between the sham and hypoxia condition. As EMG activity of these muscles was not assessed during the current study, and due to the challenges of assessing single muscle forces in humans, it is unclear if the relative contribution of these muscles to elbow flexor force was altered with fatigue.

Conclusions

The current study assessed voluntary activation of the elbow flexors and corticospinal excitability (i.e. single-pulse TMS) during and following a 10 min submaximal sustained isometric contraction performed in severe acute hypoxia. When exposed to 2 h of a severe acute hypoxic stimulus, the ability of the motor system to recover is impaired. This was demonstrated by the presence of a greater superimposed twitch and, subsequently, a lower voluntary activation when stimuli were delivered to either the motor cortex or the motor point of the elbow flexor muscles. Reductions in motor-evoked potential responses were seen during the recovery phase during hypoxia, indicating a decrease in the responsiveness of the corticospinal pathway. Collectively, our findings indicate that following fatiguing exercise, severe acute hypoxia impaired the supraspinal pathways responsible for muscle activation.

References

Adam A & De Luca CJ (2003). Recruitment order of motor units in human vastus lateralis muscle is maintained during fatiguing contractions. *J Neurophysiol* **90**, 2919–2927.

Adam A & De Luca CJ (2005). Firing rates of motor units in human vastus lateralis muscle during fatiguing isometric contractions. *J Appl Physiol (1985)* **99**, 268–280.

Allen D, Kabbara AA & Westerblad H(2002). Muscle fatigue: the role of intracellular calcium stores. *Can J Appl Physiol* **27**, 83–96.

Allen GM, McKenzie DK & Gandevia SC (1998). Twitch interpolation of the elbow flexor muscles at high forces. *Muscle Nerve* **21**, 318–328.

Amann M, Eldridge MW, Lovering AT, Stickland MK, Pegelow DF & Dempsey JA (2006). Arterial oxygenation influences central motor output and exercise performance via effects on peripheral locomotor muscle fatigue in humans. *J Physiol* **575**, 937–952.

Barcroft H & Millen JL (1939). The blood flow through muscle during sustained contraction. *J Physiol* **97**, 17–31.

Bigland-Ritchie B, Dawson NJ, Johansson RS & Lippold OC (1986). Reflex origin for the slowing of motoneurone firing rates in fatigue of human voluntary contractions. *J Physiol* **379**, 451–459.

Bigland-Ritchie B, Johansson R, Lippold OC, Smith S & Woods JJ (1983). Changes in motoneurone firing rates during sustained maximal voluntary contractions. *J Physiol* **340**, 335–346.

Borg GA (1982). Psychophysical bases of perceived exertion. *Med Sci Sports Exerc* **14**, 377–381.

Boroojerdi B, Battaglia F, Muellbacher W & Cohen LG (2001). Mechanisms influencing stimulus–response properties of the human corticospinal system. *Clin Neurophysiol* **112**, 931–937.

Bouillard K, Nordez A, Hodges PW, Cornu C & Hug F (2012). Evidence of changes in load sharing during isometric elbow flexion with ramped torque. *J Biomech* **45**, 1424–1429.

De Luca CJ & Contessa P (2012). Hierarchical control of motor units in voluntary contractions. *J Neurophysiol* **107**, 178–195.

De Luca CJ, LeFever RS, McCue MP & Xenakis AP (1982). Behaviour of human motor units in different muscles during linearly varying contractions. *J Physiol* **329**, 113–128.

de Morree HM, Klein C & Marcora SM (2012). Perception of effort reflects central motor command during movement execution. *Psychophysiology* **49**, 1242–1253.

Di Lazzaro V, Restuccia D, Oliviero A, Profice P, Ferrara L, Insola A, Mazzone P, Tonali P & Rothwell JC (1998). Effects of voluntary contraction on descending volleys evoked by transcranial stimulation in conscious humans. *J Physiol* **508**, 625–633.

Dousset E, Steinberg JG, Balon N & Jammes Y (2001). Effects of acute hypoxemia on force and surface EMG during sustained handgrip. *Muscle Nerve* **24**, 364–371.

Enoka RM & Stuart DG (1992). Neurobiology of muscle fatigue. *J Appl Physiol (1985)* **72**, 1631–1648.

Froyd C, Millet GY & Noakes TD (2013). The development of peripheral fatigue and short-term recovery during self-paced high-intensity exercise. *J Physiol* **591**, 1339–1346.

Fuhr P, Agostino R & Hallett M (1991). Spinal motor neuron excitability during the silent period after cortical stimulation. *Electroencephalogr Clin Neurophysiol* **81**, 257–262.

Fulco CS, Lewis SF, Frykman PN, Boushel R, Smith S, Harman EA, Cymerman A & Pandolf KB (1996). Muscle fatigue and exhaustion during dynamic leg exercise in normoxia and hypobaric hypoxia. *J Appl Physiol (1985)* **81**, 1891–1900.

Gandevia SC, Allen GM, Butler JE & Taylor JL (1996). Supraspinal factors in human muscle fatigue: evidence for suboptimal output from the motor cortex. *J Physiol* **490**, 529–536.

Goodall S, Gonzalez-Alonso J, Ali L, Ross EZ & Romer LM (2012). Supraspinal fatigue after normoxic and hypoxic exercise in humans. *J Physiol* **590**, 2767–2782.

Goodall S, Ross EZ & Romer LM (2010). Effect of graded hypoxia on supraspinal contributions to fatigue with unilateral knee-extensor contractions. *J Appl Physiol (1985)* **109**, 1842–1851.

Goodall S, Twomey R, Amann M, Ross EZ, Lovering AT, Romer LM, Subudhi AW & Roach RC (2014). AltitudeOmics: exercise-induced supraspinal fatigue is attenuated in healthy humans after acclimatization to high altitude. *Acta Physiol (Oxf)* **210**, 875–888.

Heroux ME, Butler AA, Gandevia SC, Taylor JL & Butler JE (2016). Time course of human motoneuron recovery after sustained low-level voluntary activity. *J Neurophysiol* **115**, 803–812.

Hunter SK, Ryan DL, Ortega JD & Enoka RM (2002). Task differences with the same load torque alter the endurance time of submaximal fatiguing contractions in humans. *J Neurophysiol* **88**, 3087–3096.

Jacunski M & Rafferty GF (2020). The effects of hypoxia and fatigue on skeletal muscle electromechanical delay. *Exp Physiol* **105**, 842–851.

Katayama K, Amann M, Pegelow DF, Jacques AJ & Dempsey JA (2007). Effect of arterial oxygenation on quadriceps fatigability during isolated muscle exercise. *Am J Physiol Regul Integr Comp Physiol* **292**, R1279–R1286.

Katayama K, Yoshitake Y, Watanabe K, Akima h & Ishida K (2010). Muscle deoxygenation during sustained and intermittent isometric exercise in hypoxia. *Med Sci Sports Exerc* **42**, 1269–1278.

Kavanagh JJ, McFarland AJ & Taylor JL (2019). Enhanced availability of serotonin increases activation of unfatigued muscle but exacerbates central fatigue during prolonged sustained contractions. *J Physiol* **597**, 319–332.

McKeown D, Simmonds MJ & Kavanagh JJ (2019). Reduced blood oxygen levels induce changes in low threshold motor unit firing that align with the individual's tolerance to hypoxia. *J Neurophysiol* **121**, 1664–1671.

McKeown DJ, McNeil CJ, Simmonds MJ & Kavanagh JJ (2020). Time course of neuromuscular responses to acute hypoxia during voluntary contractions. *Exp Physiol* **105**, 1855–1868.

McNeil CJ, Allen MD, Olympico E, Shoemaker JK & Rice CL (2015). Blood flow and muscle oxygenation during low, moderate, and maximal sustained isometric contractions. *Am J Physiol Regul Integr Comp Physiol* **309**, R475–R481.

McNeil CJ, Giesebrecht S, Gandevia SC & Taylor JL (2011*a*). Behaviour of the motoneurone pool in a fatiguing submaximal contraction. *J Physiol* **589**, 3533–3544.

McNeil CJ, Giesebrecht S, Khan SI, Gandevia SC & Taylor JL (2011*b*). The reduction in human motoneurone responsiveness during muscle fatigue is not prevented by increased muscle spindle discharge. *J Physiol* **589**, 3731–3738.

Merton PA (1954). Voluntary strength and fatigue. *J Physiol* **123**, 553–564.

Millet GY, Muthalib M, Jubeau M, Laursen PB & Nosaka K (2012). Severe hypoxia affects exercise performance independently of afferent feedback and peripheral fatigue. *J Appl Physiol (1985)* **112**, 1335–1344.

Miscio G, Milano E, Aguilar J, Savia G, Foffani G, Mauro A, Mordillo-Mateos L, Romero-Ganuza J & Oliviero A (2009). Functional involvement of central nervous system at high altitude. *Exp Brain Res* **194**, 157–162.

Murray WM, Buchanan TS & Delp SL (2002). Scaling of peak moment arms of elbow muscles with upper extremity bone dimensions. *J Biomech* **35**, 19–26.

Rojas-Camayo J, Mejia CR, Callacondo D, Dawson JA, Posso M, Galvan CA, Davila-Arango N, Bravo EA, Loescher VY, Padilla-Deza MM, Rojas-Valero N, Velasquez-Chavez G, Clemente J, Alva-Lozada G, Quispe-Mauricio A, Bardalez S & Subhi R (2018). Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. *Thorax* **73**, 776–778.

Romer LM, Haverkamp HC, Lovering AT, Pegelow DF & Dempsey JA (2006). Effect of exercise-induced arterial hypoxemia on quadriceps muscle fatigue in healthy humans. *Am J Physiol Regul Integr Comp Physiol* **290**, R365–R375.

Ruggiero L, Yacyshyn AF, Nettleton J & McNeil CJ (2018). UBC-Nepal expedition: acclimatization to high-altitude increases spinal motoneurone excitability during fatigue in humans. *J Physiol* **596**, 3327–3339.

Rupp T, Jubeau M, Wuyam B, Perrey S, Levy P, Millet GY & Verges S (2012). Time-dependent effect of acute hypoxia on corticospinal excitability in healthy humans. *J Neurophysiol* **108**, 1270–1277.

Rupp T, Racinais S, Bringard A, Lapole T & Perrey S (2015). Modulation of exercise-induced spinal loop properties in response to oxygen availability. *Eur J Appl Physiol* **115**, 471–482.

Sacco P, Thickbroom GW, Thompson ML & Mastaglia FL (1997). Changes in corticomotor excitation and inhibition during prolonged submaximal muscle contractions. *Muscle Nerve* **20**, 1158–1166.

Sogaard K, Gandevia SC, Todd G, Petersen NT & Taylor JL (2006). The effect of sustained low-intensity contractions on supraspinal fatigue in human elbow flexor muscles. *J Physiol* **573**, 511–523.

Staudenmann D, Rudroff T & Enoka RM (2009). Pronation-supination torque and associated electromyographic activity varies during a sustained elbow flexor contraction but does not influence the time to task failure. *Muscle Nerve* **40**, 231–239.

Szubski C, Burtscher M & Loscher WN (2006). The effects of short-term hypoxia on motor cortex excitability and neuromuscular activation. *J Appl Physiol (1985)* **101**, 1673–1677.

Szubski C, Burtscher M & Loscher WN (2007). Neuromuscular fatigue during sustained contractions performed in short-term hypoxia. *Med Sci Sports Exerc* **39**, 948–954.

Taylor JL, Butler JE, Allen GM & Gandevia SC (1996). Changes in motor cortical excitability during human muscle fatigue. *J Physiol* **490**, 519–528.

Todd G, Taylor JL & Gandevia SC (2003). Measurement of voluntary activation of fresh and fatigued human muscles using transcranial magnetic stimulation. *J Physiol* **551**, 661–671.

Todd G, Taylor JL & Gandevia SC (2004). Reproducible measurement of voluntary activation of human elbow flexors with motor cortical stimulation. *J Appl Physiol (1985)* **97**, 236–242.

Todd G, Taylor JL & Gandevia SC (2016). Measurement of voluntary activation based on transcranial magnetic stimulation over the motor cortex. *J Appl Physiol (1985)* **121**, 678–686.

Townsend N, Brocherie F, Millet GP & Girard O (2021). Central and peripheral muscle fatigue following repeated-sprint running in moderate and severe hypoxia. *Exp Physiol* **106**, 126–138.

Additional Information

Data availability statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Competing interest

The authors declare that they have no competing interests.

Author contributions

All authors contributed to the conception and design of this work, as well as the drafting and final approval of the manuscript. All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. All persons designated as authors qualify for authorship, and all those who qualify for authorship are listed. Data collection and analysis were performed by D.J.M. and E.J.B. at Griffith University, Australia.

Funding

The authors declare that no funding was obtained for this work.

Acknowledgements

The authors would like to express their gratitude to all the participants involved in our study, for their support and patience, and for contributing their time.

Keywords

corticospinal excitability, deoxygenation, fatigue, peripheral nerve stimulation, transcranial magnetic stimulation

Supporting information

Additional supporting information can be found online in the Supporting Information section at the end of the HTML view of the article. Supporting information files available:

Peer Review History
Statistical Summary Document